

Muaaz Bhamjee

Associate Professor, Department of Mechanical and Aeronautical Engineering
University of Pretoria

Email: muaaz.bhamjee@up.ac.za

List of Publications

JOURNAL PUBLICATIONS

(Accredited Journals. ATLAS/CERN collaboration papers listed below as indexed on Google Scholar.)

Non-ATLAS Journal Papers

1. Khanyisani Mhlangano Makhanya and Muaaz Bhamjee and Neil Martinson and Simon Connell, “Assessment and characterization of the impact of pulmonary pathology on flow-induced acoustics using computational fluid dynamics”, *Biomedical Signal Processing and Control*, vol. 120, pp. 110018, 2026. [DOI](#)
2. Ralijaona, Mbolahasina and Igumbor, Emmanuel and Bhamjee, Muaaz and Otjombe, Kennedy and Nabeemeeah, Firdaus and Milovanovich, Minja and Martinson, Neil and Mafa, Pedro and Leeuw, Lerethodi and Connell, Simon, “Infectiousness model of expelled droplets exposed to ultraviolet germicidal irradiation coupled with evaporation”, *Computers & Fluids*, vol. 275, pp. 106242, 2024. [DOI](#)
3. Mokoena, Fortune and Bhamjee, Muaaz and Connell, Simon H. and Van Vaerenbergh, Pierre and Iori, Gianluca and Kaprolat, Axel, “An FEA Investigation of the Vibration Response of the BEATS Detector Stage”, *R&D Journal of the South African Institution of Mechanical Engineering*, vol. 39, pp. 44–52, 2023. [DOI](#)
4. Makhanya, K. M. and Connell, S. H. and Bhamjee, Muaaz and Martinson, N. A., “The Use of Computational Fluid Dynamics for Assessing Flow-Induced Acoustics to Diagnose Lung Conditions”, *Mathematical and Computational Applications*, vol. 28, No. 3, pp. 64, 2023. [DOI](#)
5. Bhamjee, Muaaz and Connell, Simon H. and Nel, André Leon, “The Modification of the Dynamic Behaviour of the Cyclonic Flow in a Hydrocyclone under Surging Conditions”, *Mathematical and Computational Applications*, vol. 27, No. 6, pp. 88, 2022. [DOI](#)
6. Potgieter, A. and Bhamjee, M. and Kruger, S., “Modelling of a Heated Gas-solid Fluidised Bed using Eulerian Based Models”, *R&D Journal of the South African Institution of Mechanical Engineering*, vol. 37, pp. 45–57, 2021. [DOI](#)
7. Potgieter, M. S. W. and Bester, C. R. and Bhamjee, M., “Experimental and CFD Investigation of a Hybrid Solar Air Heater”, *Solar Energy*, vol. 195, pp. 413–428, 2020. Impact Factor: 5.742. [DOI](#)
8. Coetzee, Rigardt Alfred Maarten and Jen, Tien-Chien and Bhamjee, Muaaz and Lu, Junling, “The mechanistic process analysis and temperature effect in a low pressure square type atomic layer deposition reactor”, *International Journal of Mechanical Engineering and Robotics Research*, vol. 8, No. 5, pp. 792–796, 2019. [DOI](#)
9. Coetzee, Rigardt Alfred Maarten and Jen, Tien-Chien and Bhamjee, Muaaz and Lu, Junling, “The mechanistic effect over the substrate in a square type atomic layer deposition reactor”, *International Journal of Modern Physics B*, vol. 33, No. 01n03, pp. 1940018, 2019. [DOI](#)
10. Bhamjee, M. and Nurick, A. and Madyira, D. M., “An Experimentally Validated Mathematical and CFD Model of a Supply Air Window: Forced and Natural Flow”, *Energy and Buildings*, vol. 57, pp. 289–301, 2013. Impact Factor: 5.879. [DOI](#)

ATLAS/CERN Collaboration Papers

Note: Papers on which Bhamjee is a contributing author, as indexed on Google Scholar.

1. ATLAS Collaboration, “Search for boosted low-mass resonances decaying into hadrons produced in association with a photon in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”,

- Journal of High Energy Physics*, vol. 2025, No. 1, pp. 99, 2025.
2. ATLAS Collaboration, “Measurements of W^+W^- production cross-sections in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2025, No. 8, pp. 1–66, 2025.
 3. ATLAS Collaboration, “The performance of missing transverse momentum reconstruction and its significance with the ATLAS detector using 140 fb^{-1} of $\sqrt{s} = 13$ TeV pp collisions”, *The European Physical Journal C*, vol. 85, No. 6, pp. 606, 2025.
 4. ATLAS Collaboration, “Differential cross-section measurements of Higgs boson production in the $H \rightarrow \tau^+\tau^-$ decay channel in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2025, No. 3, pp. 1–66, 2025.
 5. ATLAS Collaboration, “Search for diphoton resonances in the 66 to 110 GeV mass range using pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2025, No. 1, pp. 53, 2025. DOI
 6. ATLAS Collaboration, “Observation of Υ Production in Pb+Pb Collisions at $\sqrt{s_{\text{NN}}}$ with the ATLAS Detector”, *Physical Review Letters*, vol. 134, No. 14, 2025.
 7. ATLAS Collaboration, “Exploration at the high-energy frontier: ATLAS Run 2 searches investigating the exotic jungle beyond the Standard Model”, *Physics Reports*, vol. 1116, pp. 301–385, 2025.
 8. ATLAS Collaboration, “The quest to discover supersymmetry at the ATLAS experiment”, *Physics Reports*, vol. 1116, pp. 261–300, 2025.
 9. ATLAS Collaboration, “Characterising the Higgs boson with ATLAS data from the LHC Run-2”, *Physics Reports*, vol. 1116, pp. 4–56, 2025.
 10. ATLAS Collaboration, “Climbing to the Top of the ATLAS 13 TeV data”, *Physics Reports*, vol. 1116, pp. 127–183, 2025.
 11. ATLAS Collaboration, “Software and computing for Run 3 of the ATLAS experiment at the LHC”, *The European Physical Journal C*, vol. 85, No. 3, pp. 234, 2025.
 12. ATLAS Collaboration, “A search for R-parity-violating supersymmetry in final states containing many jets in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2024, No. 5, pp. 003, 2024.
 13. ATLAS Collaboration, “A search for top-squark pair production, in final states containing a top quark, a charm quark and missing transverse momentum, using the 139 fb^{-1} of pp collision data collected by the ATLAS detector”, *Journal of High Energy Physics*, vol. 2024, No. 7, pp. 1–49, 2024.
 14. ATLAS Collaboration, “Combination of measurements of the top quark mass from data collected by the ATLAS and CMS experiments at $\sqrt{s} = 7$ and 8 TeV”, *Physical Review Letters*, vol. 132, No. 26, pp. 261902, 2024.
 15. ATLAS Collaboration, “Evidence for the $VH, H \rightarrow \tau\tau$ process with the ATLAS detector in Run 2”, *Physics Letters B*, vol. 855, pp. 138817, 2024.
 16. ATLAS Collaboration, “Fiducial and differential cross-section measurements of electroweak $W\gamma jj$ production in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *The European Physical Journal C*, vol. 84, No. 10, pp. 1064, 2024.
 17. ATLAS Collaboration, “Improving topological cluster reconstruction using calorimeter cell timing in ATLAS”, *The European Physical Journal C*, vol. 84, No. 5, pp. 455, 2024.
 18. ATLAS Collaboration, “Measurements of inclusive and differential cross-sections of $t\bar{t}\gamma$ production in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2024, No. 10, pp. 191, 2024.
 19. ATLAS Collaboration, “Operation and performance of the ATLAS tile calorimeter in LHC Run 2”, *The European Physical Journal C*, vol. 84, No. 12, pp. 1313, 2024.
 20. ATLAS Collaboration, “Precise measurements of W^- and Z -boson transverse momentum spectra with the ATLAS detector using pp collisions at $\sqrt{s} = 5.02$ TeV and 13 TeV”, *The European Physical Journal C*, vol. 84, No. 10, pp. 1126, 2024.
 21. ATLAS Collaboration, “Search for nearly mass-degenerate higgsinos using low-momentum mildly displaced tracks in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Physical*

- Review Letters*, vol. 132, No. 22, pp. 221801, 2024.
22. ATLAS Collaboration, “Search for a new Z' gauge boson via the $pp \rightarrow W^{\pm(*)} \rightarrow Z'\mu^{\pm}\nu \rightarrow \mu^{\pm}\mu^{\mp}\mu^{\pm}\nu$ process in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Physical Review D*, vol. 110, No. 7, pp. 072008, 2024.
 23. ATLAS Collaboration, “Search for a resonance decaying into a scalar particle and a Higgs boson in the final state with two bottom quarks and two photons in proton–proton collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2024, No. 11, pp. 1–47, 2024.
 24. ATLAS Collaboration, “Search for charged-lepton-flavor violating $\mu\tau qt$ interactions in top-quark production and decay in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector at the LHC”, *Physical Review D*, vol. 110, No. 1, pp. 012014, 2024.
 25. ATLAS Collaboration, “Search for light long-lived neutral particles from Higgs boson decays via vector-boson-fusion production from pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *The European Physical Journal C*, vol. 84, No. 7, pp. 719, 2024.
 26. ATLAS Collaboration, “Search for new particles in final states with a boosted top quark and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2024, No. 5, pp. 263, 2024.
 27. ATLAS Collaboration, “Search for non-resonant production of semi-visible jets using Run 2 data in ATLAS”, *Physics Letters B*, vol. 848, pp. 138324, 2024.
 28. ATLAS Collaboration, “Search for pair production of higgsinos in events with two Higgs bosons and missing transverse momentum in $\sqrt{s} = 13$ TeV pp collisions at the ATLAS experiment”, *Physical Review D*, vol. 109, No. 11, pp. 112011, 2024.
 29. ATLAS Collaboration, “Search for singly produced vectorlike top partners in multilepton final states with 139 fb $^{-1}$ of pp collision data at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Physical Review D*, vol. 109, No. 11, pp. 032005, 2024.
 30. ATLAS Collaboration, “Search for the decay of the Higgs boson to a Z boson and a light pseudoscalar particle decaying to two photons”, *Physics Letters B*, vol. 850, pp. 138536, 2024.
 31. ATLAS Collaboration, “Search for the non-resonant production of Higgs boson pairs via gluon fusion and vector-boson fusion in the $b\bar{b}\tau^+\tau^-$ final state in proton–proton collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Physical Review D*, vol. 110, No. 3, pp. 032012, 2024.
 32. ATLAS Collaboration, “Search for top-philic heavy resonances in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *The European Physical Journal C*, vol. 84, No. 2, pp. 157, 2024.
 33. ATLAS Collaboration, “Statistical combination of ATLAS Run 2 searches for charginos and neutralinos at the LHC”, *Physical Review Letters*, vol. 133, No. 3, pp. 031802, 2024.
 34. ATLAS Collaboration, “Studies of the energy dependence of diboson polarization fractions and the radiation-amplitude-zero effect in WZ production with the ATLAS detector”, *Physical Review Letters*, vol. 133, No. 10, pp. 101802, 2024.
 35. ATLAS Collaboration, “Study of $Z \rightarrow \ell\ell\gamma$ decays at $\sqrt{s} = 8$ TeV with the ATLAS detector”, *The European Physical Journal C*, vol. 84, No. 195, pp. 195, 2024.
 36. ATLAS Collaboration, “Underlying-event studies with strange hadrons in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *The European Physical Journal C*, vol. 84, No. 12, pp. 1335, 2024.
 37. ATLAS Collaboration, “Measurement of jet substructure in boosted $t\bar{t}$ events with the ATLAS detector using 140 fb $^{-1}$ of 13 TeV pp collisions”, *Physical Review D*, vol. 109, No. 11, pp. 112016, 2024.
 38. ATLAS Collaboration, “Measurement of the $t\bar{t}$ cross section and its ratio to the Z production cross section using pp collisions at $\sqrt{s} = 13.6$ TeV with the ATLAS detector”, *Physics Letters B*, vol. 848, pp. 138376, 2024.
 39. ATLAS Collaboration, “Search for new phenomena in two-body invariant mass distributions using unsupervised machine learning for anomaly detection at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Physical Review Letters*, vol. 132, No. 8, pp. 081801, 2024.
 40. ATLAS Collaboration, “Search for non-resonant Higgs boson pair production in the $2b + 2\ell + E_{\text{T}}^{\text{miss}}$ final state in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Journal of High*

- Energy Physics*, vol. 2024, No. 2, pp. 037, 2024.
41. ATLAS Collaboration, “Search for pair production of squarks or gluinos decaying via sleptons or weak bosons in final states with two same-sign or three leptons with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2024, No. 2, pp. 107, 2024.
 42. ATLAS Collaboration, “Search for the $Z\gamma$ decay mode of new high-mass resonances in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Physics Letters B*, vol. 848, pp. 138394, 2024.
 43. ATLAS Collaboration, “Observation of quantum entanglement with top quarks at the ATLAS detector”, *Nature*, vol. 633, No. 8030, pp. 542–547, 2024.
 44. ATLAS Collaboration, “Observation of $W\gamma\gamma$ triboson production in proton-proton collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Physics Letters B*, vol. 848, pp. 138400, 2024.
 45. ATLAS Collaboration, “Observation of electroweak production of W^+W^- in association with jets in proton-proton collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2024, No. 7, 2024.
 46. ATLAS Collaboration, “Combination of searches for Higgs boson pair production in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Physical Review Letters*, vol. 133, No. 10, pp. 101801, 2024.
 47. ATLAS Collaboration, “Determination of the relative sign of the Higgs boson couplings to W and Z bosons using WH production via vector-boson fusion with the ATLAS detector”, *Physical Review Letters*, vol. 133, No. 14, pp. 141801, 2024.
 48. ATLAS Collaboration, “Combination and summary of ATLAS dark matter searches interpreted in a 2HDM with a pseudo-scalar mediator using 139 fb^{-1} of $\sqrt{s} = 13$ TeV pp collision data”, *Science Bulletin*, vol. 69, No. 19, pp. 3005–3035, 2024.
 49. ATLAS Collaboration, “The ATLAS experiment at the CERN Large Hadron Collider: a description of the detector configuration for Run 3”, *Journal of Instrumentation*, vol. 19, No. 05, pp. P05063, 2024.
 50. ATLAS Collaboration, “The ATLAS trigger system for LHC Run 3 and trigger performance in 2022”, *Journal of Instrumentation*, vol. 19, No. 06, pp. P06029, 2024.
 51. ATLAS Collaboration, “Measurement of ZZ production cross-sections in the four-lepton final state in pp collisions at $\sqrt{s} = 13.6$ TeV with the ATLAS experiment”, *Physics Letters B*, vol. 855, pp. 138764, 2024.
 52. ATLAS Collaboration, “Measurement and interpretation of same-sign W boson pair production in association with two jets in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2024, No. 4, 2024.
 53. ATLAS Collaboration, “Measurement of the Z boson invisible width at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Physics Letters B*, vol. 854, pp. 138705, 2024.
 54. ATLAS Collaboration, “Inclusive and differential cross-section measurements of $t\bar{t}Z$ production in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector, including EFT and spin-correlation interpretations”, *Journal of High Energy Physics*, vol. 2024, No. 7, 2024.
 55. ATLAS Collaboration, “Electron and photon efficiencies in LHC Run 2 with the ATLAS experiment”, *Journal of High Energy Physics*, vol. 2024, No. 5, 2024.
 56. ATLAS Collaboration, “Measurement of t -channel production of single top quarks and anti-quarks in pp collisions at 13 TeV using the full ATLAS Run 2 data sample”, *Journal of High Energy Physics*, vol. 2024, No. 5, 2024.
 57. ATLAS Collaboration, “Software performance of the ATLAS track reconstruction for LHC Run 3”, *Computing and Software for Big Science*, vol. 8, No. 1, pp. 9, 2024.
 58. ATLAS Collaboration, “A precise measurement of the Z -boson double-differential transverse momentum and rapidity distributions in the full phase space of the decay leptons with the ATLAS experiment at $\sqrt{s} = 8$ TeV”, *The European Physical Journal C*, vol. 84, No. 3, pp. 315, 2024.
 59. ATLAS Collaboration, “Measurement of the production cross-section of J/ψ and $\psi(2S)$ mesons in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *The European Physical Journal C*, vol. 84, No. 2, pp. 169, 2024.

60. ATLAS Collaboration, “Search for resonant production of dark quarks in the dijet final state with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2024, No. 2, 2024.
61. ATLAS Collaboration, “Studies of new Higgs boson interactions through nonresonant HH production in the $b\bar{b}\gamma\gamma$ final state in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2024, No. 1, pp. 66, 2024.
62. ATLAS Collaboration, “Differential cross-section measurements of the production of four charged leptons in association with two jets using the ATLAS detector”, *Journal of High Energy Physics*, vol. 2024, No. 1, pp. 4, 2024.
63. ATLAS Collaboration, “Simultaneous energy and mass calibration of large-radius jets with the ATLAS detector using a deep neural network”, *Machine Learning: Science and Technology*, vol. 5, No. 3, pp. 035051, 2024.
64. ATLAS Collaboration, “Search for new phenomena with top-quark pairs and large missing transverse momentum using 140 fb $^{-1}$ of pp collision data at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2024, No. 3, 2024.
65. ATLAS Collaboration, “Measurement of the centrality dependence of the dijet yield in p +Pb collisions at $\sqrt{s_{NN}} = 8.16$ TeV with the ATLAS Detector”, *Physical Review Letters*, vol. 132, No. 10, 2024.
66. ATLAS Collaboration, “Search for a CP-odd Higgs boson decaying into a heavy CP-even Higgs boson and a Z boson in the $\ell^+\ell^-t\bar{t}$ and $\nu\bar{\nu}b\bar{b}$ final states using 140 fb $^{-1}$ of data collected with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2024, No. 2, 2024.
67. ATLAS Collaboration, “Search for the Exclusive W Boson Hadronic Decays $W^\pm \rightarrow \pi^\pm\gamma$, $W^\pm \rightarrow K^\pm\gamma$ and $W^\pm \rightarrow \rho^\pm\gamma$ with the ATLAS Detector”, *Physical Review Letters*, vol. 133, No. 16, pp. 161804, 2024.
68. ATLAS Collaboration, “Measurement of the W -boson mass and width with the ATLAS detector using proton–proton collisions at $\sqrt{s} = 7$ TeV”, *The European Physical Journal C*, vol. 84, No. 12, pp. 1309, 2024.
69. ATLAS Collaboration, “Interpretations of the ATLAS measurements of Higgs boson production and decay rates and differential cross-sections in pp collisions at $\sqrt{s} = 13$ TeV”, *Journal of High Energy Physics*, vol. 2024, No. 11, pp. 97, 2024.
70. ATLAS Collaboration, “Combination of searches for heavy spin-1 resonances using 139 fb $^{-1}$ of proton-proton collision data at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2024, No. 4, 2024.
71. ATLAS Collaboration, “Search for pair-production of vector-like quarks in lepton+jets final states containing at least one b -tagged jet using the Run 2 data from the ATLAS experiment”, *Physics Letters B*, vol. 854, pp. 138743, 2024.
72. ATLAS Collaboration, “Combination of searches for pair-produced leptoquarks at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Physics Letters B*, vol. 854, pp. 138736, 2024.
73. ATLAS Collaboration, “Search for short- and long-lived axion-like particles in $H \rightarrow aa \rightarrow 4\gamma$ decays with the ATLAS experiment at the LHC”, *The European Physical Journal C*, vol. 84, No. 7, pp. 742, 2024.
74. ATLAS Collaboration, “Combination of searches for resonant Higgs boson pair production using pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Physical Review Letters*, vol. 132, No. 23, pp. 231801, 2024.
75. ATLAS Collaboration, “Electron and photon energy calibration with the ATLAS detector using LHC Run 2 data”, *Journal of Instrumentation*, vol. 19, No. 02, pp. P02009, 2024.
76. ATLAS Collaboration, “Search for electroweak production of supersymmetric particles in final states with two τ -leptons in $\sqrt{s} = 13$ TeV pp collisions with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2024, No. 5, 2024.
77. ATLAS Collaboration, “Performance and calibration of quark/gluon-jet taggers using 140 fb $^{-1}$ of pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Chinese Physics C*, vol. 48, No. 2, pp. 023001, 2024.
78. ATLAS Collaboration, “Probing the CP nature of the top–Higgs Yukawa coupling in $t\bar{t}H$ and tH events with $H \rightarrow b\bar{b}$ decays using the ATLAS detector at the LHC”, *Physics Letters B*,

- vol. 849, pp. 138469, 2024.
79. ATLAS Collaboration, “Search for pair-produced higgsinos decaying via Higgs or Z bosons to final states containing a pair of photons and a pair of b -jets with the ATLAS detector”, *Physics Letters B*, vol. 856, pp. 138938, 2024.
 80. ATLAS Collaboration, “Measurement of the $H \rightarrow \gamma\gamma$ and $H \rightarrow ZZ^* \rightarrow 4\ell$ cross-sections in pp collisions at $\sqrt{s} = 13.6$ TeV with the ATLAS detector”, *The European Physical Journal C*, vol. 84, No. 1, pp. 78, 2024.
 81. ATLAS Collaboration, “Study of high-transverse-momentum Higgs boson production in association with a vector boson in the $qqb\bar{b}$ final state with the ATLAS detector”, *Physical Review Letters*, vol. 132, 2024.
 82. ATLAS Collaboration, “Test of CP-invariance of the Higgs boson in vector-boson fusion production and in its decay into four leptons”, *Journal of High Energy Physics*, vol. 2024, No. 5, 2024.
 83. ATLAS Collaboration, “Search for heavy resonances in final states with four leptons and missing transverse momentum or jets in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2024, No. 10, 2024.
 84. ATLAS Collaboration, “Measurement of differential cross-sections in $t\bar{t}$ and $t\bar{t}$ +jets production in the lepton+jets final state in pp collisions at $\sqrt{s} = 13$ TeV using 140 fb $^{-1}$ of ATLAS data”, *Journal of High Energy Physics*, vol. 2024, No. 8, 2024.
 85. ATLAS Collaboration, “Differential cross-sections for events with missing transverse momentum and jets measured with the ATLAS detector in 13 TeV proton-proton collisions”, *Journal of High Energy Physics*, vol. 2024, No. 8, 2024.
 86. ATLAS Collaboration, “Measurement of the VH , $H \rightarrow \tau\tau$ process with the ATLAS detector at 13 TeV”, *Physics Letters B*, vol. 855, pp. 138817, 2024.
 87. ATLAS Collaboration, “Measurements of Lund subjet multiplicities in 13 TeV proton-proton collisions with the ATLAS detector”, *Physics Letters B*, vol. 859, pp. 139090, 2024.
 88. ATLAS Collaboration, “Measurements of jet cross-section ratios in 13 TeV proton-proton collisions with ATLAS”, *Physical Review D*, vol. 110, No. 7, pp. 072019, 2024.
 89. ATLAS Collaboration, “Search for low-mass resonances decaying into two jets and produced in association with a photon or a jet at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Physical Review D*, vol. 110, No. 3, pp. 032002, 2024.
 90. ATLAS Collaboration, “Fast b -tagging at the high-level trigger of the ATLAS experiment in LHC Run 3”, *Journal of Instrumentation*, vol. 18, No. 11, pp. P11006, 2023.
 91. ATLAS Collaboration, “A search for heavy Higgs bosons decaying into vector bosons in same-sign two-lepton final states in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2023, No. 7, pp. 200, 2023.
 92. ATLAS Collaboration, “Anomaly detection search for new resonances decaying into a Higgs boson and a generic new particle X in hadronic final states using $\sqrt{s} = 13$ TeV pp collisions with the ATLAS detector”, *Physical Review D*, vol. 108, No. 5, pp. 052009, 2023.
 93. ATLAS Collaboration, “Charged-hadron production in pp , p +Pb, Pb+Pb, and Xe+Xe collisions at $\sqrt{s_{NN}} = 5$ TeV with the ATLAS detector at the LHC”, *Journal of High Energy Physics*, vol. 2023, No. 7, pp. 74, 2023.
 94. ATLAS Collaboration, “Combination of searches for invisible decays of the Higgs boson using 139 fb $^{-1}$ of proton-proton collision data at $\sqrt{s} = 13$ TeV collected with the ATLAS experiment”, *Physics Letters B*, vol. 842, pp. 137963, 2023.
 95. ATLAS Collaboration, “Constraints on spin-0 dark matter mediators and invisible Higgs decays using ATLAS 13 TeV pp collision data with two top quarks and missing transverse momentum in the final state”, *The European Physical Journal C*, vol. 83, No. 6, pp. 503, 2023.
 96. ATLAS Collaboration, “Cross-section measurements for the production of a Z boson in association with high-transverse-momentum jets in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2023, No. 6, pp. 80, 2023.
 97. ATLAS Collaboration, “Determination of the strong coupling constant from transverse energy-energy correlations in multijet events at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Journal of*

- High Energy Physics*, vol. 2023, No. 7, pp. 85, 2023.
98. ATLAS Collaboration, “Exclusive dielectron production in ultraperipheral Pb+Pb collisions at $\sqrt{s_{NN}} = 5.02$ TeV with ATLAS”, *Journal of High Energy Physics*, vol. 2023, No. 6, pp. 182, 2023.
 99. ATLAS Collaboration, “Fiducial and differential cross-section measurements for the vector-boson-fusion production of the Higgs boson in the $H \rightarrow WW^* \rightarrow e\nu\mu\nu$ decay channel at 13 TeV with the ATLAS detector”, *Physical Review D*, vol. 108, No. 7, pp. 072003, 2023.
 100. ATLAS Collaboration, “Inclusive and differential cross-sections for dilepton $t\bar{t}$ production measured in $\sqrt{s} = 13$ TeV pp collisions with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2023, No. 7, pp. 141, 2023.
 101. ATLAS Collaboration, “Inclusive-photon production and its dependence on photon isolation in pp collisions at $\sqrt{s} = 13$ TeV using 139 fb^{-1} of ATLAS data”, *Journal of High Energy Physics*, vol. 2023, No. 7, pp. 86, 2023.
 102. ATLAS Collaboration, “Measurement of suppression of large-radius jets and its dependence on substructure in Pb+Pb collisions at $\sqrt{s_{NN}} = 5.02$ TeV with the ATLAS detector”, *Physical Review Letters*, vol. 131, No. 17, pp. 172301, 2023.
 103. ATLAS Collaboration, “Measurement of electroweak $Z(\nu\bar{\nu})\gamma jj$ production and limits on anomalous quartic gauge couplings in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2023, No. 6, pp. 82, 2023.
 104. ATLAS Collaboration, “Measurement of exclusive pion pair production in proton–proton collisions at $\sqrt{s} = 7$ TeV with the ATLAS detector”, *The European Physical Journal C*, vol. 83, No. 7, pp. 627, 2023.
 105. ATLAS Collaboration, “Measurement of single top-quark production in the s-channel in proton–proton collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2023, No. 6, pp. 191, 2023.
 106. ATLAS Collaboration, “Measurement of the $B_s^0 \rightarrow \mu\mu$ effective lifetime with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2023, No. 9, pp. 1–32, 2023.
 107. ATLAS Collaboration, “Measurement of the sensitivity of two-particle correlations in pp collisions to the presence of hard scatterings”, *Physical Review Letters*, vol. 131, pp. 162301, 2023.
 108. ATLAS Collaboration, “Measurement of the charge asymmetry in top-quark pair production in association with a photon with the ATLAS experiment”, *Physics Letters B*, vol. 843, pp. 137848, 2023.
 109. ATLAS Collaboration, “Measurement of the cross-sections of the electroweak and total production of a $Z\gamma$ pair in association with two jets in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Physics Letters B*, vol. 846, pp. 138222, 2023.
 110. ATLAS Collaboration, “Measurement of the nuclear modification factor of b-jets in 5.02 TeV Pb+Pb collisions with the ATLAS detector”, *The European Physical Journal C*, vol. 83, No. 5, pp. 438, 2023.
 111. ATLAS Collaboration, “Measurement of the polarisation of W bosons produced in top-quark decays using dilepton”, *Physics Letters B*, vol. 843, pp. 137829, 2023.
 112. ATLAS Collaboration, “Measurement of the production of a W boson in association with a charmed hadron in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Physical Review D*, vol. 108, No. 3, pp. 032012, 2023.
 113. ATLAS Collaboration, “Measurement of the properties of Higgs boson production at $\sqrt{s} = 13$ TeV in the $H \rightarrow \gamma\gamma$ channel using 139 fb^{-1} of pp collision data with the ATLAS experiment”, *Journal of High Energy Physics*, vol. 2023, No. 7, pp. 88, 2023.
 114. ATLAS Collaboration, “Measurement of the total cross section and ρ -parameter from elastic scattering in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *The European Physical Journal C*, vol. 83, No. 5, pp. 441, 2023.
 115. ATLAS Collaboration, “Measurement of the CP properties of Higgs boson interactions with τ -leptons with the ATLAS detector”, *The European Physical Journal C*, vol. 83, No. 7, pp. 563, 2023.

116. ATLAS Collaboration, “Measurements of W^+W^- production in decay topologies inspired by searches for electroweak supersymmetry”, *The European Physical Journal C*, vol. 83, No. 8, pp. 718, 2023.
117. ATLAS Collaboration, “Measurements of differential cross sections of Higgs boson production through gluon fusion in the $H \rightarrow WW^* \rightarrow e\nu\mu\nu$ final state at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *The European Physical Journal C*, vol. 83, No. 9, pp. 774, 2023.
118. ATLAS Collaboration, “Measurements of multijet event isotropies using optimal transport with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2023, No. 10, pp. 1–58, 2023.
119. ATLAS Collaboration, “Measurements of observables sensitive to colour reconnection in $t\bar{t}$ events with the ATLAS detector at $\sqrt{s} = 13$ TeV”, *The European Physical Journal C*, vol. 83, No. 6, pp. 518, 2023.
120. ATLAS Collaboration, “Model-independent search for the presence of new physics in events including $H \rightarrow \gamma\gamma$ with $\sqrt{s} = 13$ TeV pp data recorded by the ATLAS detector at the LHC”, *Journal of High Energy Physics*, vol. 2023, No. 7, pp. 176, 2023.
121. ATLAS Collaboration, “Search for $t\bar{t}H/A \rightarrow t\bar{t}t\bar{t}$ production in the multilepton final state in proton–proton collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2023, No. 7, pp. 203, 2023.
122. ATLAS Collaboration, “Search for Majorana neutrinos in same-sign WW scattering events from pp collisions at $\sqrt{s} = 13$ TeV”, *The European Physical Journal C*, vol. 83, No. 9, pp. 824, 2023.
123. ATLAS Collaboration, “Search for a new Z' gauge boson in 4μ events with the ATLAS experiment”, *Journal of High Energy Physics*, vol. 2023, No. 7, pp. 90, 2023.
124. ATLAS Collaboration, “Search for a new pseudoscalar decaying into a pair of muons in events with a top-quark pair at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Physical Review D*, vol. 108, No. 9, pp. 092007, 2023.
125. ATLAS Collaboration, “Search for a new scalar resonance in flavour-changing neutral-current top-quark decays $t \rightarrow qX$ ($q = u, c$), with $X \rightarrow b\bar{b}$, in proton–proton collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2023, No. 7, pp. 199, 2023.
126. ATLAS Collaboration, “Search for an axion-like particle with forward proton scattering in association with photon pairs at ATLAS”, *Journal of High Energy Physics*, vol. 2023, No. 7, pp. 234, 2023.
127. ATLAS Collaboration, “Search for dark matter produced in association with a single top quark and an energetic W boson in $\sqrt{s} = 13$ TeV pp collisions with the ATLAS detector”, *The European Physical Journal C*, vol. 83, No. 7, pp. 603, 2023.
128. ATLAS Collaboration, “Search for dark matter produced in association with a Higgs boson decaying to tau leptons at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2023, No. 9, pp. 189, 2023.
129. ATLAS Collaboration, “Search for direct production of electroweakinos in final states with one lepton, jets and missing transverse momentum in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2023, No. 12, pp. 167, 2023.
130. ATLAS Collaboration, “Search for direct production of winos and higgsinos in events with two same-charge leptons or three leptons in pp collision data at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2023, No. 11, pp. 150, 2023.
131. ATLAS Collaboration, “Search for doubly charged Higgs boson production in multi-lepton final states using 139 fb^{-1} of proton–proton collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *The European Physical Journal C*, vol. 83, No. 7, pp. 605, 2023.
132. ATLAS Collaboration, “Search for excited τ -leptons and leptoquarks in the final state with τ -leptons and jets in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2023, No. 6, pp. 199, 2023.
133. ATLAS Collaboration, “Search for exclusive Higgs and Z boson decays to $\omega\gamma$ and Higgs boson decays to $K^*\gamma$ with the ATLAS detector”, *Physics Letters B*, vol. 847, pp. 138292, 2023.
134. ATLAS Collaboration, “Search for flavour-changing neutral current interactions of the top quark and the Higgs boson in events with a pair of τ – leptons in pp collisions at $\sqrt{s} = 13$ TeV

- with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2023, No. 6, pp. 155, 2023.
135. ATLAS Collaboration, “Search for flavour-changing neutral tqH interactions with $H \rightarrow \gamma\gamma$ in pp collisions at $\sqrt{s} = 13$ TeV using the ATLAS detector”, *Journal of High Energy Physics*, vol. 2023, No. 12, pp. 195, 2023.
 136. ATLAS Collaboration, “Search for heavy Majorana or Dirac neutrinos and right-handed W gauge bosons in final states with charged leptons and jets in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *The European Physical Journal C*, vol. 83, No. 12, pp. 1164, 2023.
 137. ATLAS Collaboration, “Search for heavy long-lived multi-charged particles in the full LHC Run 2 pp collision data at $\sqrt{s} = 13$ TeV using the ATLAS detector”, *Physics Letters B*, vol. 847, pp. 138316, 2023.
 138. ATLAS Collaboration, “Search for heavy resonances decaying into a Z or W boson and a Higgs boson in final states with leptons and b-jets in 139 fb⁻¹ of pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2023, No. 6, pp. 16, 2023.
 139. ATLAS Collaboration, “Search for heavy Higgs bosons with flavour-violating couplings in multi-lepton plus b-jets final states in pp collisions at 13 TeV with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2023, No. 12, pp. 81, 2023.
 140. ATLAS Collaboration, “Search for lepton-flavour violation in high-mass dilepton final states using 139 fb⁻¹ of pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2023, No. 10, pp. 082, 2023.
 141. ATLAS Collaboration, “Search for leptonic charge asymmetry in $t\bar{t}W$ production in final states with three leptons at $\sqrt{s} = 13$ TeV”, *Journal of High Energy Physics*, vol. 2023, No. 7, pp. 33, 2023.
 142. ATLAS Collaboration, “Search for leptoquarks decaying into the $b\tau$ final state in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2023, No. 10, pp. 1, 2023.
 143. ATLAS Collaboration, “Search for long-lived, massive particles in events with displaced vertices and multiple jets in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2023, No. 6, pp. 200, 2023.
 144. ATLAS Collaboration, “Search for magnetic monopoles and stable particles with high electric charges in $\sqrt{s} = 13$ TeV pp collisions with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2023, No. 11, pp. 112, 2023.
 145. ATLAS Collaboration, “Search for new phenomena in multi-body invariant masses in events with at least one isolated lepton and two jets using $\sqrt{s} = 13$ TeV proton–proton collision data collected by the ATLAS detector”, *Journal of High Energy Physics*, vol. 2023, No. 7, pp. 202, 2023.
 146. ATLAS Collaboration, “Search for nonresonant pair production of Higgs bosons in the $b\bar{b}b\bar{b}$ final state in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Physical Review D*, vol. 108, pp. 052003, 2023.
 147. ATLAS Collaboration, “Search for pair production of third-generation leptoquarks decaying into a bottom quark and a τ -lepton with the ATLAS detector”, *The European Physical Journal C*, vol. 83, No. 11, pp. 1075, 2023.
 148. ATLAS Collaboration, “Search for pair-produced scalar and vector leptoquarks decaying into third-generation quarks and first-or second-generation leptons in pp collisions with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2023, No. 6, pp. 188, 2023.
 149. ATLAS Collaboration, “Search for pair-produced vector-like top and bottom partners in events with large missing transverse momentum in pp collisions with the ATLAS detector”, *The European Physical Journal C*, vol. 83, No. 8, pp. 719, 2023.
 150. ATLAS Collaboration, “Search for pair-production of vector-like quarks in pp collision events at $\sqrt{s} = 13$ TeV with at least one leptonically decaying Z boson and a third-generation quark with the ATLAS detector”, *Physics Letters B*, vol. 843, pp. 138019, 2023.
 151. ATLAS Collaboration, “Search for pairs of muons with small displacements in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Physics Letters B*, vol. 846, pp. 138172, 2023.

152. ATLAS Collaboration, “Search for periodic signals in the dielectron and diphoton invariant mass spectra using 139 fb^{-1} of pp collisions at $\sqrt{s} = 13 \text{ TeV}$ with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2023, No. 10, pp. 1–51, 2023.
153. ATLAS Collaboration, “Search for single production of vector-like T quarks decaying into Ht or Zt in pp collisions at $\sqrt{s} = 13 \text{ TeV}$ with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2023, No. 8, pp. 153, 2023.
154. ATLAS Collaboration, “Search for third-generation vector-like leptons in pp collisions at $\sqrt{s} = 13 \text{ TeV}$ with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2023, No. 7, pp. 118, 2023.
155. ATLAS Collaboration, “Search for vector-boson resonances decaying into a top quark and a bottom quark using pp collisions at $\sqrt{s} = 13 \text{ TeV}$ with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2023, No. 12, pp. 073, 2023.
156. ATLAS Collaboration, “Search for Higgs boson pair production in association with a vector boson in pp collisions at $\sqrt{s} = 13 \text{ TeV}$ with the ATLAS detector”, *The European Physical Journal C*, vol. 83, No. 6, pp. 519, 2023.
157. ATLAS Collaboration, “Searches for exclusive Higgs and Z boson decays into a vector quarkonium state and a photon using 139 fb^{-1} of ATLAS $\sqrt{s} = 13 \text{ TeV}$ proton–proton collision data”, *The European Physical Journal C*, vol. 83, No. 9, pp. 781, 2023.
158. ATLAS Collaboration, “Studies of the muon momentum calibration and performance of the ATLAS detector with pp collisions at $\sqrt{s} = 13 \text{ TeV}$ ”, *The European Physical Journal C*, vol. 83, No. 8, pp. 686, 2023.
159. ATLAS Collaboration, “A search for new resonances in multiple final states with a high transverse momentum Z boson in $\sqrt{s} = 13 \text{ TeV}$ pp collisions with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2023, No. 6, pp. 036, 2023.
160. ATLAS Collaboration, “Search for dark matter produced in association with a dark Higgs boson decaying into W^+W^- in the one-lepton final state at $\sqrt{s} = 13 \text{ TeV}$ using 139 fb^{-1} of pp collisions recorded with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2023, No. 7, pp. 116, 2023.
161. ATLAS Collaboration, “Search for dark photons from Higgs boson decays via ZH production with a photon plus missing transverse momentum signature from pp collisions at $\sqrt{s} = 13 \text{ TeV}$ with the ATLAS detector”, *Journal of High Energy Physics*, vol. 2023, No. 7, pp. 133, 2023.
162. ATLAS Collaboration, “Search for resonant $WZ \rightarrow \ell\nu\ell'\ell'$ production in proton–proton collisions at $\sqrt{s} = 13 \text{ TeV}$ with the ATLAS detector”, *The European Physical Journal C*, vol. 83, pp. 633, 2023.
163. ATLAS Collaboration, “Search for supersymmetry in final states with missing transverse momentum and three or more b-jets in 139 fb^{-1} of proton–proton collisions at $\sqrt{s} = 13 \text{ TeV}$ with the ATLAS detector”, *The European Physical Journal C*, vol. 83, pp. 561, 2023.
164. ATLAS Collaboration, “Strong constraints on jet quenching in centrality-dependent p+Pb collisions at 5.02 TeV from ATLAS”, *Physical Review Letters*, vol. 131, No. 7, pp. 072301, 2023.
165. ATLAS Collaboration, “Study of the CP property of the Higgs boson to electroweak boson coupling in the VBF $H \rightarrow \gamma\gamma$ channel with the ATLAS detector”, *Physical Review Letters*, vol. 131, No. 6, pp. 061802, 2023.
166. ATLAS Collaboration, “Correlations between flow and transverse momentum in Xe+Xe and Pb+Pb collisions at the LHC with the ATLAS detector”, *Physical Review C*, vol. 107, No. 5, pp. 054910, 2023. [DOI](#)
167. ATLAS Collaboration, “Measurement of substructure-dependent jet suppression in Pb+Pb collisions at 5.02 TeV with the ATLAS detector”, *Physical Review C*, vol. 107, No. 5, pp. 054909, 2023. [DOI](#)
168. ATLAS Collaboration, “Measurements of the suppression and correlations of dijets in Pb+Pb collisions at $\sqrt{s_{\text{NN}}} = 5.02 \text{ TeV}$ ”, *Physical Review C*, vol. 107, No. 5, pp. 054908, 2023. [DOI](#)
169. ATLAS Collaboration, “Measurement of muon pairs produced via $\gamma\gamma$ scattering in nonultra-peripheral Pb+Pb collisions at $\sqrt{s_{\text{NN}}} = 5.02 \text{ TeV}$ with the ATLAS detector”, *Physical Review*

- C*, vol. 107, No. 5, pp. 054907, 2023. [DOI](#)
170. ATLAS Collaboration, “Measurement of the total and differential Higgs boson production cross-sections at $\sqrt{s} = 13$ TeV with the ATLAS detector by combining the $H \rightarrow ZZ^* \rightarrow 4\ell$ and $H \rightarrow \gamma\gamma$ decay channels”, *Journal of High Energy Physics*, vol. 2023, No. 5, pp. 28, 2023. [DOI](#)
 171. ATLAS Collaboration, “Observation of four-top-quark production in the multilepton final state with the ATLAS detector”, *The European Physical Journal C*, vol. 83, No. 6, pp. 496, 2023.
 172. ATLAS Collaboration, “Production of $\Upsilon(nS)$ mesons in Pb+Pb and pp collisions at 5.02 TeV”, *Physical Review C*, vol. 107, No. 5, pp. 054912, 2023.
 173. ATLAS Collaboration, “Combined measurement of the Higgs boson mass from the $H \rightarrow \gamma\gamma$ and $H \rightarrow ZZ^* \rightarrow 4\ell$ decay channels with the ATLAS detector using $\sqrt{s} = 7, 8$, and 13 TeV pp collision data”, *Physical Review Letters*, vol. 131, No. 25, 2023.
 174. ATLAS Collaboration, “Observation of single-top-quark production in association with a photon using the ATLAS detector”, *Physical Review Letters*, vol. 131, 2023.
 175. ATLAS Collaboration, “Observation of gauge boson joint-polarisation states in $W^\pm Z$ production from pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Physics Letters B*, vol. 843, pp. 137895, 2023.
 176. ATLAS Collaboration, “Evidence of off-shell Higgs boson production from ZZ leptonic decay channels and constraints on its total width with the ATLAS detector”, *Physics Letters B*, vol. 846, pp. 138223, 2023.
 177. ATLAS Collaboration, “Observation of the $\gamma\gamma \rightarrow \tau\tau$ process in Pb+Pb collisions and constraints on the τ -lepton anomalous magnetic moment with the ATLAS detector”, *Physical Review Letters*, vol. 131, No. 15, 2023.
 178. ATLAS Collaboration, “Test of CP invariance in Higgs boson vector-boson-fusion production using the $H \rightarrow \gamma\gamma$ channel with the ATLAS detector”, *Physical Review Letters*, vol. 131, No. 6, 2023.
 179. ATLAS Collaboration, “Constraints on the Higgs boson self-coupling from single- and double-Higgs production with the ATLAS detector using pp collisions at $\sqrt{s} = 13$ TeV”, *Physics Letters B*, vol. 843, pp. 137745, 2023.
 180. ATLAS Collaboration, “Measurement of the Higgs boson mass with $H \rightarrow \gamma\gamma$ decays in 140 fb^{-1} of $\sqrt{s} = 13$ TeV pp collisions with the ATLAS detector”, *Physics Letters B*, vol. 847, pp. 138315, 2023.
 181. ATLAS Collaboration, “Comparison of inclusive and photon-tagged jet suppression in 5.02 TeV Pb+Pb collisions with ATLAS”, *Physics Letters B*, vol. 846, pp. 138154, 2023.
 182. ATLAS Collaboration, “Measurements of the suppression and correlations of dijets in Xe+Xe collisions at $\sqrt{s_{\text{NN}}} = 5.44$ TeV”, *Physical Review C*, vol. 108, No. 2, 2023.
 183. ATLAS Collaboration, “New techniques for jet calibration with the ATLAS detector”, *The European Physical Journal C*, vol. 83, No. 8, pp. 761, 2023.
 184. ATLAS Collaboration, “Calibration of the light-flavour jet mistagging efficiency of the b -tagging algorithms with Z +jets events using 139 fb^{-1} of ATLAS proton–proton collision data at $\sqrt{s} = 13$ TeV”, *The European Physical Journal C*, vol. 83, No. 8, pp. 728, 2023.
 185. ATLAS Collaboration, “Measurement of the Higgs boson mass in the $H \rightarrow ZZ^* \rightarrow 4\ell$ decay channel using 139 fb^{-1} of $\sqrt{s} = 13$ TeV pp collisions recorded by the ATLAS detector at the LHC”, *Physics Letters B*, vol. 843, pp. 137880, 2023.
 186. ATLAS Collaboration, “Tools for estimating fake/non-prompt lepton backgrounds with the ATLAS detector at the LHC”, *Journal of Instrumentation*, vol. 18, No. 11, pp. T11004, 2023.
 187. ATLAS Collaboration, “Performance of the reconstruction of large impact parameter tracks in the inner detector of ATLAS”, *The European Physical Journal C*, vol. 83, No. 11, pp. 1081, 2023.
 188. ATLAS Collaboration, “Evidence of pair production of longitudinally polarised vector bosons and study of CP properties in $ZZ \rightarrow 4\ell$ events with the ATLAS detector at $\sqrt{s} = 13$ TeV”, *Journal of High Energy Physics*, vol. 2023, No. 12, pp. 107, 2023.

189. ATLAS Collaboration, “Search for resonant WZ production in the fully leptonic final state in proton–proton collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Physics Letters B*, vol. 787, pp. 68–88, 2018. [DOI](#)

PATENTS

1. “Dynamically Forecasting High Resolution Air Temperature in Real-Time Using Multiple Sources”

Inventors: Akhalwaya, Ismail Yunus and Vos, Etienne Eben and Makhanya, Sibusisiwe Audrey and Govindasamy, Tamara Rosemary and Patel, Zubeida and Morake, Dedricks Monyai and Mahlasi, Craig and Bhamjee, Muaaz.

Inventors on behalf of IBM Corp. South Africa: Granted 30 July 2024, ZA2023/09448 (ID: 99151468). USA: Pending, US18/200,050. [Link](#)

BOOK CHAPTERS

1. Rashied, Naiefa and Bhamjee, Muaaz, “Does the Global South Need to Decolonise the Fourth Industrial Revolution?”, in *The Disruptive Fourth Industrial Revolution: Technology, Society and Beyond*, eds. Doorsamy, W. and Paul, B. and Marwala, T., Lecture Notes in Electrical Engineering vol. 674, pp. 95–110, Springer International Publishing, Cham, 2020. [DOI](#)

CONFERENCE PUBLICATIONS

Published

1. Mngomezulu, Mangaliso and Gaffoor, Zaheed and Vos, Etienne and Govindasamy, Tamara and Mahlasi, Craig and Bhamjee, Muaaz and Baloyi, Gciniwe and Kuehnert, Julian and Mwangi, Francis and Kadima, Cleophas and Makhanya, Sibusisiwe, “Auxiliary Feature Injection For Prithvi-ViT Fine-Tuning”, *IGARSS 2025 – 2025 IEEE International Geoscience and Remote Sensing Symposium*, Brisbane, Australia, pp. 1075–1078, 2025.
2. Abbajee, Taheer Jooma and Anderson, Keegan Doig and Bhamjee, Muaaz and Visaya, Maria Vivien, “Reservoir Computing for Predicting Chaotic Dynamical Systems”, *69th Annual Conference of the South African Institute of Physics (SAIP 2025)*, pp. 1026–1031, 2025.
3. Cassim, Abdool Sattar and Igumbor, Emmanuel and Connell, Simon Henry and Rudolph, Michael and Kgolobe, Rethabile and Harley, Charis and Bhamjee, Muaaz and Brooks, Tim, “IoT-Based Environmental Conditions Monitoring of a Sawtooth Greenhouse: A Foundation for Anomaly Detection and Computational Fluid Dynamics”, *69th Annual Conference of the South African Institute of Physics (SAIP 2025)*, pp. 818–823, 2025.
4. Baloyi, Gciniwe Simphiwe and Bhamjee, Muaaz and Gaffoor, Zaheed and Govindasamy, Tamara Rosemary and Mahlasi, Craig and Makhanya, Sibusisiwe and Mngomezulu, Mangaliso and Vos, Etienne Eben, “IBM EOFM Enabling Seamless Characterization of Surface Urban Heat Islands (SUHIs)”, *Climate and Health Africa Conference*, 2024. [Link](#)
5. Bhamjee, Muaaz and Gaffoor, Zaheed and Govindasamy, Tamara Rosemary and Mahlasi, Craig and Makhanya, Sibusisiwe and Mngomezulu, Mangaliso and Vos, Etienne Eben, “Foundation model technology for Urban Heat Islands”, *Deep Learning Indaba 2024*, 2024. [Link](#)
6. Bhamjee, Muaaz and Debary, Hiyam and Gaffoor, Zaheed and Govindasamy, Tamara and Mahlasi, Craig and Fiaz, Mustansar and Vos, Etienne and Klein, Levente and Makhanya, Sibusisiwe and Watson, Campbell, “Detection and Characterization of Urban Heat Islands with Machine Learning”, *IGARSS 2024 – 2024 IEEE International Geoscience and Remote Sensing Symposium*, Athens, Greece, pp. 1693–1699, 2024. [DOI](#)
7. Igumbor, E. and Ralijaona, M. and Bhamjee, M. and Otwombe, K. and Nabeemeeah, F. and Martinson, N. and Mafa Takisa, P. and Leeuw, L. and Connell, S. H., “Modeling the infectiousness of droplets when exposed to ultra-violet germicidal system: A computational fluid dynamics approach”, *Annual Conference of the South African Institute of Physics (SAIP 2022)*, pp. 689–694, 2022.
8. Ralijaona, M. and Igumbor, E. and Bhamjee, M. and Connell, S. H. and Martinson, N. and

- Otwombe, K. and Grobler, H. and Swanepoel, F. and Mafa Takisa, P. and Chewparsad, J., "Developing an infectiousness model for droplet transmission", *Annual Conference of the South African Institute of Physics (SAIP 2022)*, pp. 594–599, 2022.
9. Connell, Matthew P. and Bhamjee, Muaaz and Mafa Takisa, Pedro and Connell, Simon H. and Leeuw, Lerothodi, "Environmental Monitoring in the ATLAS ITk Detector", *Annual Conference of the South African Institute of Physics (SAIP 2022)*, pp. 553–558, 2022.
 10. Simpson, Zach and Bhamjee, Muaaz, "First-Year Mechanical Engineering Students' Strategies Around Integrating Text and Image in Design Reports", *2022 IEEE International Professional Communication Conference (ProComm)*, Limerick, Ireland, pp. 234–240, 2022. [DOI](#)
 11. Ramokoka, Tshiamo and Bhamjee, Muaaz, "Modelling fluid and particulate flow through a ventriculoperitoneal shunt in a variable temperature environment", *12th South African Conference on Computational and Applied Mechanics (SACAM2020)*, pp. 00035, 2021. [DOI](#)
 12. Bhamjee, Muaaz and Simpson, Zach, "Online engineering tutorials: Investigation into tutors' use of technologies for learning", *2021 World Engineering Education Forum / Global Engineering Deans Council (WEEF/GEDC)*, Madrid, Spain, pp. 50–58, 2021. [DOI](#)
 13. Simpson, Zach and Bhamjee, Muaaz, "Voice in first-year engineering design report writing: An academic literacies investigation", *Proceedings of REES AAEE 2021 Conference*, Perth, Australia, 2021. [Link](#)
 14. Coetzee, Rigardt Alfred Maarten and Hoenselaar, Damon James and Bhamjee, Muaaz and Jen, Tien-Chien, "The Topology Within Recirculating Flow in the Atomic Layer Deposition Thin Film Process", *ASME International Mechanical Engineering Congress and Exposition (IMECE 2020)*, pp. V02AT02A029, 2020. [DOI](#)
 15. Hoenselaar, Damon James and Coetzee, Rigardt Alfred Maarten and Bhamjee, Muaaz and Jen, Tien-Chien, "The Flow Evolution of the Atomic Layer Deposition Process: A Numerical Study of the Implementation of a Porous Plate", *2020 3rd International Conference on Power and Energy Applications (ICPEA)*, Busan, South Korea, pp. 52–57, 2020. [DOI](#)
 16. Molale, T. and Ahmed, N. and Bhamjee, M., "Numerical Analysis of Flow Induced Vibrations of a Low-Pressure Steam Turbine Rotating Blade", *IOP: Journal of Physics: Conference Series*, pp. 032059, 2019.
 17. Simpson, Zach and Bhamjee, Muaaz, "Design Literacy Practices in a Mechanical Engineering Degree Program: A Multimodal Social Semiotic Analysis of First and Final Year Design Reports", *8th Research in Engineering Education Symposium (REES 2019)*, Cape Town, pp. 520–529, 2019.
 18. Bhamjee, Muaaz and Rashied, Naiefa, "A guide to online assessment in large engineering design classrooms", *South Africa International Conference on Education (SAICed 2018)*, Pretoria, pp. 395–406, 2018.
 19. Chirnside, David Mark and Bhamjee, Muaaz, "A numerically stable Eulerian-Eulerian model of air-core formation in a hydrocyclone", *11th South African Conference on Computational and Applied Mechanics (SACAM 2018)*, Vanderbijlpark, pp. 322–333, 2018.
 20. Chirnside, David Mark and Bhamjee, Muaaz, "A Numerically Stable Eulerian-Eulerian Model of Air-Core Formation in a Hydrocyclone", *11th South African Conference on Computational and Applied Mechanics (SACAM 2018)*, Vanderbijlpark, 2018. [Link](#)
 21. Bhamjee, M., "Investigation of the effect of lift and drag on air-core formation in a hydrocyclone", *IFSA 2017, Industrial Fluidization South Africa*, South Africa, pp. 215–225, 2017.
 22. Simpson, Zach and Bhamjee, Muaaz, "The Literacy of Engineering Design: Investigation into First Year Design Report Writing", *SASEE 2017*, Cape Town, pp. 294, 2017.
 23. Tina, S. A. Tcheuhebou and Bhamjee, Muaaz and Nel, Andre L., "Design Life Cycle of a 3-D Printed Hydrocyclone", *International Conference on Competitive Manufacturing (COMA 2016)*, Stellenbosch, pp. 301–306, 2016.
 24. Evans-Tokaryk, Tyler and Bhamjee, Muaaz and Simpson, Zachary, "Establishing a Benchmark for Effective Intervention: First-Year Engineering Students' Writing and Their Perceptions Thereof", *3rd Biennial Conference of the South African Society for Engineering Education (SASEE 2015)*, Durban, pp. 114–126, 2015.

25. Cook, M. and Tchonang, M. and Chinaka, E. and Bhamjee, M. and Bornman, F. and Connell, S. H., “Overview of the Mineral-PET run-of-mine Diamond bearing rock sorter”, *SAIP 2014, 59th Annual Conference of the South African Institute of Physics*, Johannesburg, pp. 479–484, 2014.
26. Bhamjee, Muaaz and Connell, Simon H. and Nel, Andre L., “An Investigation into the Applicability of the Lattice Boltzmann Method to Modelling of the Flow in a Hydrocyclone”, *9th South African Conference on Computational and Applied Mechanics (SACAM 2014)*, Somerset West, pp. SACAM_020, 2014.
27. Bhamjee, Muaaz and Connell, Simon H. and Nel, Andre L., “The Effect of Surface Tension on Air-core Formation in a Hydrocyclone”, *9th South African Conference on Computational and Applied Mechanics (SACAM 2014)*, Somerset West, pp. SACAM_021, 2014.
28. Potgieter, Martin S. W. and Nurick, Alan and Bhamjee, Muaaz, “A Computational and Experimental Analysis of a Hybrid Parallel/Counter Flow Double Pass Solar Air Heater”, *9th South African Conference on Computational and Applied Mechanics (SACAM 2014)*, Somerset West, pp. SACAM_054, 2014.
29. Bhamjee, Muaaz and Nurick, Alan, “The Finite Element Method and its Link to the Finite Difference Method for Poisson’s Equation”, *8th South African Conference on Computational and Applied Mechanics (SACAM 2012)*, Johannesburg, pp. 148–153, 2012.
30. Madyira, Daniel M. and Bhamjee, Muaaz, “Comparative Study of the Performance of Displacement and Conventional Ventilation using CFD”, *Proceedings of the Third IASTED African Conference (AfricaMS 2010)*, Gaborone, Botswana, pp. 056–149, 2010.

OTHER SCHOLARLY CONTRIBUTIONS

1. Aad, Georges and Aakvaag, Erlend and Abbott, Braden Keim and Abdelhameed, Sara and Abeling, Kira and Abicht, Nils Julius and Abidi, Haider and Aboelela, Mohammed and Aboulhorma, Asmaa and Abramowicz, Halina and others, “Observation of pair production of longitudinally polarized Z bosons in four-lepton final state with the ATLAS detector”, , 2026.
2. Aad, Georges and Aakvaag, Erlend and Abbott, Braden Keim and Abdelhameed, Sara and Abeling, Kira and Abicht, Nils Julius and Abidi, Haider and Aboelela, Mohammed and Aboulhorma, Asmaa and Abramowicz, Halina and others, “Search for Lorentz-boosted di- τ resonances produced in association with top quark pairs in $\sqrt{s} = 13$ TeV pp collisions with the ATLAS detector”, , 2026.
3. Aad, Georges and Aakvaag, Erlend and Abbott, Braden Keim and Abdelhameed, Sara and Abeling, Kira and Abicht, Nils Julius and Abidi, Haider and Aboelela, Mohammed and Aboulhorma, Asmaa and Abramowicz, Halina and others, “Observation of the electroweak production of $\gamma\gamma jj$ at $\sqrt{s} = 13$ TeV in 140 fb^{-1} of pp collision data with the ATLAS detector”, , 2026.
4. Aad, Georges and Aakvaag, Erlend and Abbott, Braden Keim and Abdelhameed, Sara and Abeling, Kira and Abicht, Nils Julius and Abidi, Haider and Aboelela, Mohammed and Aboulhorma, Asmaa and Abramowicz, Halina and others, “Evidence for longitudinally polarised Z bosons in electroweak $W^{\pm}Z$ production in association with two jets from pp collisions at $\sqrt{s} = 13$ TeV and 13.6 TeV with the ATLAS detector”, , 2026.
5. Aad, Georges and Aakvaag, Erlend and Abbott, Braden Keim and Abdelhameed, Sara and Abeling, Kira and Abicht, Nils Julius and Abidi, Haider and Aboelela, Mohammed and Aboulhorma, Asmaa and Abramowicz, Halina and others, “Search for new scalars via $X \rightarrow SH \rightarrow b\bar{b}b\bar{b}$ in proton-proton collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, , 2026.
6. Aad, Georges and Aakvaag, Erlend and Abbott, Braden Keim and Abdelhameed, Sara and Abeling, Kira and Abicht, Nils Julius and Abidi, Haider and Aboelela, Mohammed and Aboulhorma, Asmaa and Abramowicz, Halina and others, “Improved analysis of non-resonant Higgs boson pair production in the $b\bar{b}\tau^{+}\tau^{-}$ final state with 196 fb^{-1} of data collected at $\sqrt{s} = 13$ TeV and 13.6 TeV with the ATLAS detector”, , 2026.
7. Aad, Georges and Aakvaag, Erlend and Abbott, Braden Keim and Abdelhameed, Sara and Abeling, Kira and Abicht, Nils Julius and Abidi, Haider and Aboelela, Mohammed and Aboul-

- horma, Asmaa and Abramowicz, Halina and others, “Search for heavy Majorana neutrinos in vector boson scattering with τ -lepton final states with the ATLAS detector”, , 2026.
8. Aad, Georges and Aakvaag, Erlend and Abbott, Braden Keim and Abdelhameed, Sara and Abeling, Kira and Abicht, Nils Julius and Abidi, Haider and Aboelela, Mohammed and Aboulhorma, Asmaa and Abramowicz, Halina and others, “Search for tH production in the $H \rightarrow \tau\tau$ decay mode, and a combination with other searches, using pp collisions at 13 TeV and 13.6 TeV with the ATLAS detector”, , 2026.
 9. Aad, Georges and Aakvaag, Erlend and Abbott, Braden Keim and Abdelhameed, Sara and Abeling, Kira and Abicht, Nils Julius and Abidi, Haider and Aboelela, Mohammed and Aboulhorma, Asmaa and Abramowicz, Halina and others, “Search for high-mass dilepton resonances in pp collisions at $\sqrt{s} = 13.6$ TeV combined with 13 TeV results using the ATLAS detector”, , 2026.
 10. Aad, Georges and Aakvaag, Erlend and Abbott, Braden Keim and Abdelhameed, Sara and Abeling, Kira and Abicht, Nils Julius and Abidi, Haider and Aboelela, Mohammed and Aboulhorma, Asmaa and Abramowicz, Halina and others, “The calibration of large-radius jets using the Run 2 dataset with the ATLAS detector”, , 2026.
 11. Aad, Georges and Aakvaag, Erlend and Abbott, Braden Keim and Abdelhameed, Sara and Abeling, Kira and Abicht, Nils Julius and Abidi, Haider and Aboelela, Mohammed and Aboulhorma, Asmaa and Abramowicz, Halina and others, “Search for pair-production of vector-like T quarks decaying into a top quark and a spin-0 particle in the diphoton final state in proton proton collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, , 2026.
 12. Aad, Georges and Aakvaag, Erlend and Abbott, Braden Keim and Abdelhameed, Sara and Abeling, Kira and Abicht, Nils Julius and Abidi, Haider and Aboelela, Mohammed and Aboulhorma, Asmaa and Abramowicz, Halina and others, “Study of $t\bar{t}H$ and tH production in the $H \rightarrow \tau\tau$ channel in pp collisions at $\sqrt{s} = 13$ TeV and 13.6 TeV with the ATLAS detector”, , 2026.
 13. Bhamjee, Muaaz and Connell, Matthew and Connell, Simon and Igumbor, Emmanuel and Leeuw, Lerothodi and Mafa, Pedro and Oriunno, Marco and Vreeswijk, Marcel, “Characterisation of the Thermoflow due to the Dry Nitrogen Flushing Scheme in the ATLAS Inner Tracker using Computational Fluid Dynamics”, arXiv preprint, 2026. [Link](#)
 14. Bhamjee, Muaaz, “MKM411 SciML Lectures — Scientific Machine Learning Lecture Notebooks for Undergraduate CFD Education”, , 2026. [Link](#)
 15. Bhamjee, Muaaz and Craig, Ken J, “MKM411 PINN Heat Transfer — Physics-Informed Neural Networks for Unsteady Heat Transfer”, , 2026. [Link](#)
 16. Aad, Georges and Aakvaag, Erlend and Abbott, Braden Keim and Abdelhameed, Sara and Abeling, Kira and Abicht, Nils Julius and Abidi, Haider and Aboelela, Mohammed and Aboulhorma, Asmaa and Abramowicz, Halina and others, “Measurement of the Higgs boson decay to a low-mass dilepton system and a photon in pp collisions at $\sqrt{s} = 13$ and 13.6 TeV with the ATLAS detector”, ATLAS Collaboration paper, ATLAS-HIGP-2024-16, 2026.
 17. Aad, Georges and Aakvaag, Erlend and Abbott, Braden Keim and Abdelhameed, Sara and Abeling, Kira and Abicht, Nils Julius and Abidi, Haider and Aboelela, Mohammed and Aboulhorma, Asmaa and Abramowicz, Halina and others, “Measurement of tZq inclusive and differential cross-sections in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, ATLAS Collaboration paper, ATLAS-TOPQ-2021-21, 2026.
 18. Aad, Georges and Aakvaag, Erlend and Abbott, Braden Keim and Abdelhameed, Sara and Abeling, Kira and Abicht, Nils Julius and Abidi, Haider and Aboelela, Mohammed and Aboulhorma, Asmaa and Abramowicz, Halina and others, “Constraints on the Higgs boson total width from on-shell signal-background interference in the $H \rightarrow \gamma\gamma$ decay channel with pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, ATLAS Collaboration paper, ATLAS-HIGP-2024-12, 2026.
 19. Bhamjee, Muaaz, “Entangled Worlds: Engineering, Physics, Applied Mathematics, and the Future of High-Performance Computing”, Invited Talk, CHPC National Conference 2025, HPC Applications Track, 2025. [Link](#)

20. Bhamjee, Muaaz and Gaffoor, Zaheed and Govindasamy, Tamara and Mahlasi, Craig and Vos, Etienne and Mngomezulu, Mangaliso and Baloyi, Gciniwe and Makhanya, Sibusisiwe, “Open-Source AI Model: Fine-Tuned Earth Observation Foundation Model (EOFM) for Land Surface Temperature Prediction”, `ibm-granite/granite-geospatial-land-surface-temperature`, HuggingFace / GitHub, 2024. [Link](#)
21. Bhamjee, Muaaz and Gaffoor, Zaheed and Govindasamy, Tamara Rosemary and Mahlasi, Craig and Makhanya, Sibusisiwe and Vos, Etienne Eben, “Urban Heat Island Estimation with Machine Learning”, Poster, DS-I Africa Consortium Meeting 2023, 2023. [Link](#)
22. Aad, Georges and others, “Measurements of longitudinal flow decorrelations in pp and $Xe+Xe$ collisions with the ATLAS detector”, ATLAS Collaboration, ATLAS-HION-2022-17, 2023.
23. Aad, Georges and others, “A precise determination of the strong-coupling constant from the recoil of Z bosons with the ATLAS experiment at $\sqrt{s} = 8$ TeV”, ATLAS Collaboration, arXiv preprint, 2023.
24. Mafa Takisa, P. and Bhamjee, M. and Connell, S. H. and Leeuw, L. and Potgieter, M. S. W. and Oriunno, M., “CFD humidity and temperature modelling in the ATLAS ITK Strip”, SAIP 2022 (non-accredited), 2022.
25. Aad, Georges and others, “Evidence for the charge asymmetry in $pp \rightarrow t\bar{t}$ production at $\sqrt{s} = 13$ TeV with the ATLAS detector”, ATLAS Collaboration, arXiv preprint, 2022. [Link](#)
26. Bhamjee, Muaaz, “HPC as an Enabler for Growth in Computational Fluid Dynamics Research”, Presentation, 14th CHPC National Conference, Online, 1 December 2020, 2020. [Link](#)
27. Bhamjee, Muaaz, “Book review: Samuel, MA, Dhunpath, R & Amin, N (eds.). 2016. Disrupting higher education curriculum: undoing cognitive damage”, , 2018.
28. Bhamjee, Muaaz, “Modelling of Multiphase Flow in Process Equipment: The Trade-off between Accuracy and Computational Efficiency”, Keynote Presentation, 11th CHPC National Conference, Pretoria, 6 December 2017, 2017. [Link](#)
29. Bhamjee, Muaaz, “Modelling of the Multiphase Interactions in a Hydrocyclone using Navier-Stokes and Lattice Boltzmann based Computational Approaches”, DIng Thesis, University of Johannesburg, 2016. [Link](#)
30. Bhamjee, M. and Connell, S. H. and Nel, A. L., “The Computational Effect of Parallel Partitioning Approaches Based on Model Particle Distributions within the Parallel Partitions of a CFD-DEM Model”, CHPC National Meeting 2015, CSIR ICC, Pretoria, 30 November – 4 December 2015, 2015.
31. Bhamjee, Muaaz, “A Computational Fluid Dynamics and Experimental Investigation of an Airflow Window”, MIng Dissertation (Cum Laude), University of Johannesburg, 2011. [Link](#)